The Emperor's New Points: Continua Are Analysis, Not Physics

Abstract

Zeno left two kinds of paradox, and only one decides whether physics can be continuous. The paradox of measure, how points without width make a line with length, is answered by measure theory, and discreteness does not help: discrete points have no width either, and a grain with width brings the continuum back, so in every structure sizes come from relations. The paradoxes of motion are answered by analysis only as description, because a continuum has no next state, and causality needs one: a cause produces its effect as the next state, so in a physical continuum no state could be the first one a cause produces. This is an a priori proof in the logician's sense. Change without a cause is outside logic and reason. Physical events are therefore discrete steps, and that covers everything that happens: matter, gravity, spacetime, space itself on either view of space, and amplitudes, since a history produces each of its values in finitely many steps. Continuum mathematics stays sound as analysis, a description of many steps at once. Countable is not enough for physics, since the rationals are countable and dense with no next point; discrete physics needs steps with successors. A grid also owes its geometry: on a square grid the shortest route to a diagonal point stays longer than the straight line however fine the grid becomes.

1. The question that calculation alone does not answer

Zeno's paradoxes come in two kinds, and only one decides whether physics can be continuous. The paradox of measure asks how points without width add up to a length (Skyrms, 1983). Measure theory, the mathematics of length, area and

probability, answers it, and discreteness (being made of separate points or steps) does not help, since discrete points have no width either. The paradoxes of motion ask how anything reaches its next place. Analysis, the mathematics of limits, describes the run, but nothing in a continuum (a gapless expanse, like the line of real numbers) can produce the next step, because there is none.

A continuum cannot be physical, because nothing in a continuum can cause anything. Causality takes steps: a cause produces its effect, and producing an effect takes a step from one state to the next. A continuum has no next state, since between any two points lies a third, so in a physical continuum no state could be the first one a cause produces. Change without a cause is outside logic and reason: a law could at most describe it, nothing would produce it, and an account that allowed it could not say why it happens. Accepting it would repeat Plato's late Athenian anti-philosophy, which set the Forms outside nature: it puts something real beyond reason, and outside logic is illogical. Physical events are therefore discrete steps. Alfred North Whitehead reached the same conclusion from Zeno's paradoxes (Whitehead, 1978).

Continuum mathematics stays a sound tool of analysis, and its own workings agree. A line has length. Each individual point has zero length. Both statements hold, because collecting points into a set is a different operation from adding up lengths: the points do not cause the length. Resolving that mathematical question leaves a physical one: what structure accounts for the distances we measure?

The apparent contradiction: if every constituent contributes zero width, how can the whole have nonzero width? A familiar rule gives the question its force. Split a rod into two pieces and add their lengths; repeat the split finitely often and the rule still holds. The mistake is to extend that rule to an arbitrary uncountable decomposition, into more pieces than any endless list could contain, without checking what the sum means there.

Take two endpoints labeled 0 and 1. Neither endpoint has width, yet their distance is one unit. A continuous line specifies the distances among all its points; it does not assemble length by stacking point-widths. A discrete network can likewise use sites without width and give length to the links between them, and it has no other option: a site with a width would have a surface, and a surface is a continuum. Both accounts need distance relations.

1.1 The position advanced here

Physical events are connected by discrete relations, because a law produces each state from its immediate predecessor, and a dense continuum, with another point between any two, has no such predecessor-successor step. Continuous differential equations, which relate quantities to their rates of change, do not assume that kind of production; the physical case rests on it.

The premise is causality in its strong sense, production. A law can relate states at different times without any state producing the next; that is causation as dependence, and production is the stronger notion this paper relies on (Hall, 2004). Hall lets production run through continuous intermediates; the claim here is that it cannot, since a production with no first effect produces nothing. A differential equation such as dx/dt = u(t) is a law of that first kind: it says how each state depends on earlier inputs and is silent on what produces any state. In a continuum, any instant after a cause has another instant before it, so no state is the first one the cause produces, and the chain of production cannot begin. Continuum mathematics remains the tool for describing many steps at once. A reader who takes dependence to be all there is to causation finds no contradiction in a continuum, and for that reader the physics that follows carries the case.

Physics gives independent support. Gravity and quantum theory together limit how finely anything can be resolved: probing a distance near the Planck length,

about 10^-35 meters, takes so much energy in so small a space that it would form a black hole and hide what the probe was meant to show (Hossenfelder, 2013). A black hole's entropy, a measure of its number of possible internal states, is a quarter of its horizon area (the area of the boundary past which nothing escapes) in Planck units (Bekenstein, 1973; Hawking, 1975), and the covariant entropy bound that follows caps the entropy on the light rays leaving any surface without spreading apart, its light-sheets, by the surface's area, which for an ordinary region caps the distinguishable states it can hold (Bousso, 2002). A finite region therefore holds finitely many bits, if the bound holds in the fundamental theory, as is widely expected. A system with finite energy also passes through only finitely many distinguishable states in a finite time, since reaching a distinguishable state takes a minimum time set by the energy (Margolus and Levitin, 1998). That is not yet a lattice, a regular grid of points. A system with finitely many distinguishable states is still described with continuous amplitudes (numbers whose squared sizes give probabilities), and those amplitudes are analysis: a history produces each of its values in finitely many steps, so the values that occur form a countable set, one that can be listed in sequence (Section 4.1). The bound rules out a region carrying infinitely much information, which a fundamental continuum of physical values would allow.

Continuum mathematics can represent physical quantities without every coordinate in that representation being a physical location. Physical extension (taking up space) rests on discrete relations, and smooth geometry describes their collective behavior.

The same holds for space itself, on either view of what space is. If space is a thing that bodies move through, each move is a cause producing the body's next place, and a continuum has no next place to produce. If space is only the order of coexisting things and events, as Leibniz held, it is as discrete as the events that

generate it. Gravity, spacetime and matter fall under the same rule, since each changes by causes. What happens is the transitions, rules applied to rules; the relations exist as mathematics, and without their change nothing could be observed.

Two observations carry the case: a continuum has no next point, and adding its singleton widths does not yield its length. Both are consistent with continuum mathematics as a tool. Causality turns the first into a physical conclusion, since what has no next state cannot produce one. The bearer discussion below takes up the second, the requirement that distance have a physical account.

The view behind the paper is mathematical monism: everything that exists is ruled or random, and both are mathematical, an a priori proof. Continua and discrete structures alike exist as mathematics, and the physical question is which of them happen. Short laws grow minds in such steps from a simple start, since each unruled bit halves a description's weight, and an endless chain of steps has no last step to run back from, so time runs forward.

1.2 The bearer proposal

Whatever carries the extension we measure owes us a physical account. In the usual construction of Lebesgue measure on the real line, the standard way of assigning lengths to sets of points, interval lengths supply the starting size data. A singleton, a set with one point, has measure zero because it fits inside intervals of arbitrarily small length.

The physical proposal is to account for distances through discrete relations and their dynamics. A grain with positive extent would bring the continuum back inside itself, since an extent has a surface and a surface is a continuum. Elementary sites therefore have zero extent, and their separations must come from relations. Discreteness alone supplies no unit of length.

A defender of continuous space can make the same relational move: points have no individual width, while a metric assigns distances between them. The interval and its metric structure are a candidate bearer of extension. "Nothing bears extension" does not follow from "every singleton has zero length."

A stronger premise would require every extended physical region to decompose into elementary parts whose nonnegative individual extents sum to the region's extent. Together with a decomposition into zero-extent points, that premise contradicts positive extension. The inference is valid conditional on the premise, which imposes precisely the additive composition that the continuum model does not assert. The composition argument alone does not decide between the two accounts; causality decides, and the discrete account then owes a construction: specify elementary relations, derive measured geometry, and recover quantum and gravitational predictions from the same dynamics.

2. A union is not a sum of lengths

Standard measure theory handles zero-width points consistently, and its definitions show how. Stated precisely, take the usual real line with Lebesgue measure, written λ. Then

$$[0, 1] = \bigcup_{x \in [0, 1]} \{x\}, \qquad \lambda(\{x\}) = 0, \qquad \lambda([0, 1]) = 1.$$

The first equality says which points belong to the interval. It supplies no rule equating the measure of that union (all the points gathered into one set) with an uncountable sum of singleton measures.

Adding the point lengths gives a well defined answer, zero. In symbols, for any family of nonnegative numbers the generalized sum is the supremum, the least upper bound, of all finite subsums:

$$\sum_{x \in [0,\,1]} \lambda(\{x\}) = \sup_{F \subseteq [0,\,1],\ F \text{ finite}} \sum_{x \in F} 0 = 0.$$

The measure of this union is not the sum of those measures. Lebesgue measure is countably additive: for a sequence of disjoint (non-overlapping) measurable sets, the measure of the union equals the sum. The singletons of a real interval cannot be placed in such a sequence. John Hunter's lecture notes give the standard measure-theoretic treatment (Hunter, 2011).

Singleton values also fail to determine the measure. Double Lebesgue measure assigns zero to every singleton and two to the whole interval. List the points with their individual measures and you have still not said which of those two measures you have.

2.1 Ordered, but with no next point

No real position has a next one. In symbols, for any real positions a<b, the midpoint c=(a+b)/2 satisfies a<c<b, so nothing sits immediately after a. The rational numbers also have a rational midpoint between any two distinct members, despite being countable. Dense order and uncountability are different properties.

A step-by-step process can nonetheless produce a dense order. Run through the integer ratios and every positive rational appears at a finite stage. Each output has a next output in that process, although no rational has an immediate successor in numerical order. Motion can work the same way. A body at x is next at x + 3, or x + 3.25, as the result of one step, with no place in between visited, so the places a body can occupy may be dense in numerical order while every step it takes has a successor. No sequence can enumerate all real numbers, so a physical process confined to countably many successive outputs cannot put out every real position as a separate output. That confinement is an additional

premise. Numerical density alone does not deliver it.

The continuum model does not claim to step from one real number to the next. A continuous trajectory is not defined by advancing to each real number's immediate successor: x(t)=vt assigns a position to each time without requiring a next instant, and a block account, which takes a whole history as given at once, need not enumerate its positions. The discrete account of what exists, defended here, earns its place through its physical structure and consequences.

2.2 Can motion cross a continuum?

Inside the continuum model, motion crosses an interval with no arithmetic trouble. In symbols, with the speed set to one, let x(t)=t for 0<=t<=1. Every intermediate time has a corresponding position. For every nonzero time increment, Δx/Δt=1; the limit yields the same velocity with no division of zero by zero.

No step of this division produces a zero to add. Cut the duration into successive intervals of lengths 1/2, 1/4, 1/8 and so on. The first n lengths sum to 1-2^(-n), and their series (the sum carried on without end) sums to one. These are positive durations, not zeros. The continuum model therefore represents motion consistently. It does not establish that nature instantiates the model.

Bertrand Russell met the demand for a next place by holding that motion is nothing more than being at different places at different times (Russell, 1903). That pairs each time with a place, and the pairing would exist as mathematics whether or not anything moved, as a frozen record does. It describes the result of motion and leaves out what produces it: these are the emperor's new points. Zeno's runner puts the causal question (Aristotle, 1984, *Physics* VI.9). Before the finish he must reach the halfway mark, and before that the quarter mark, so no first stretch exists for him to run. Analysis solves the paradox in the abstract: x(t)=t assigns every position without a first stretch, and the stretches' times add up to

one. Production cannot proceed that way, since a cause must produce a first effect and the regress leaves none. Whitehead, arguing by Zeno's method, put it this way: once anything becomes, there is "no continuity of becoming" (Whitehead, 1978, p. 35). Steps solve the paradox in the only way causality and spacetime can happen: each step produces the next, and finitely many reach the finish.

Zeno aimed his other two paradoxes at steps. In the arrow, at each indivisible now the arrow fills a space equal to itself and so is at rest, and if time is made of nows, it does not move at all. In the stadium, two rows passing each other at one unit per smallest time pass two units of each other in that time, so passing one would take half the smallest time (Aristotle, 1984, *Physics* VI.9, 239b5-240a18). Both assume that motion happens inside a moment or passes through every place in between. A step does neither. Motion is the step itself, one state producing the next, and a body at x is next at x + 3 with no place in between visited, so no half step is owed. A snapshot holds no motion, as a frame of film holds none; the step between snapshots does. Aristotle saw where steps lead, that across a smallest part a body would have moved without ever being in the middle of moving (*Physics* VI.1, 232a), and he took it for an absurdity. It is the answer.

Two more facts finish the measure side. A countable set of points has total length zero however densely it fills an interval, so refining a grid adds no length. Discrete weights on finitely many points can still reproduce a continuum's averages within a stated bound for observables (measurable quantities) that change no faster than a fixed rate, which is how a discrete structure can underlie smooth predictions without any point carrying width.

3. Specification, construction and physical realization

A staged process cannot deliver uncountably many events. Suppose every elementary spacetime event arrives at one of a succession of stages indexed by the natural numbers, with finitely many new events at each stage. The events so generated are then at most countable, and if they exhaust spacetime's physical constituents, that premise rules out an uncountable continuum of elementary events. The argument rests on that premise. It derives no minimum spacing and excludes no continuous mathematical representation.

This countability premise is stronger than the starting-input requirement that a history generated one event at a time needs a first event. A beginning alone does not establish natural-number stages or finitely many events per stage: a continuum model may also have an initial boundary.

"Construct the interval" can mean several things. In classical mathematics, a definition can pin down a set without listing its members. A sequential enumeration cannot exhaust the reals, though inability to enumerate an object is not a contradiction in its definition. Constructive mathematics asks which objects and operations an explicit construction can deliver; identifying it with finitely many points would erase the constructive treatment of real numbers.

A circle specified by $x^2+y^2=r^2$ is exact within Euclidean geometry. A manufactured circular object must be assessed at some tolerance. No workshop reaches infinite precision, and that limit neither refutes the equation nor proves that the object inhabits discrete space.

A block or a sphere is continuous whatever carries it, since its surface is a continuum, so discreteness supplies no sizes or shapes; relations do. A block can still exist as a rule without being built. By the Minkowski-Weyl theorem, a bounded convex polytope, a solid with flat faces and no dents, is both the convex hull of finitely many points (the smallest dent-free solid containing them) and the set satisfying finitely many linear inequalities, each confining points to one side of a

plane (Ziegler, 1995, Theorem 1.1). A block's eight corners, or its six face inequalities, fix every point of its surface without listing one. Whether a step from x to x' enters the block is then decided from the two endpoints and the inequalities in finitely many operations, as in the line clipping of computer graphics, which trims lines at a region's boundary (Cyrus and Beck, 1978). The surface acts only through steps: a step that would cross it produces an event at a definite point, such as the step's rejection, and no point of the surface is built.

The same holds for curved boundaries, with one limit. Whether a point lies exactly on a perfect sphere can need every digit of its coordinates: given real numbers digit by digit, a strict inequality (a claim that one number is less than another) is confirmed at some finite stage when it holds, while equality is confirmed at none (Weihrauch, 2000). A rule that needed that decision could not finish a step. A rule that acts on definite values does not need it, so a continuous boundary acts at the precision the states carry, and its continuum stays in the description. A continuous field is a specification of the same kind. Its formula can fix how strongly, in which direction and with what amplitude each transition acts, while only the transitions occur, and computing it to any chosen precision takes finitely many steps each time, so no step needs every digit.

4. The physical question survives the mathematical answer

I reject fundamental continua and propose that continuum geometry represents very fine discrete relations. Smoothness belongs to an effective description, an approximation valid at larger scales, while the underlying structure supplies the interactions from which measured distances emerge. A continuum calculation that works is adequate for the observations it covers. It does not show that physical extension divides without limit. Three claims must be separated: a continuum is mathematically specified; a continuum model is empirically successful in a regime

(a range of conditions); a continuum is fundamental in nature. Each requires its own support.

4.1 Measure, geometry and the existing debate

The existing debate already separates the mathematical resolution from the physical question. Brian Skyrms's "Zeno's Paradox of Measure" (1983) explicitly identifies Adolf Grünbaum's work on the standard resolution using measure theory. He then asks which further physical premises would carry us from that mathematical resolution to a claim about what space is.

Baruch Spinoza, in *Ethics* I.15, rejects composing a line from points, a surface from lines or a solid from surfaces. His conclusion is not discrete space: he defends extended substance as infinite, one and indivisible. The historical lesson: rejecting a sum of zero widths does not, by itself, choose between discrete relations and a continuous whole.

Measure theory also marks where analysis outruns anything a process of steps can do. The axiom of choice lets one member be picked from each of any collection of nonempty sets, with no rule for picking. With it, some sets of points have no volume, and a ball can be cut into five pieces and reassembled, by rotations and shifts, into two balls each the size of the first (Banach and Tarski, 1924). A process of steps makes finitely many picks per step, so an endless run makes countably many, one after another, which is the axiom of dependent choice. With only that, it is consistent, given an inaccessible cardinal, that every set of reals is Lebesgue measurable (Solovay, 1970). Choice and its negation are each consistent with the other axioms (Gödel, 1938; Cohen, 1963), so both describe structures. The doubled ball exists as analysis, and no sequence of steps can perform it.

A finer grid can preserve the wrong distance. On a square lattice of spacing h with only horizontal and vertical steps, the shortest path from (0,0) to (a,a) has length 2a when a/h is an integer. The Euclidean distance (the straight-line distance) is sqrt(2)*a. Their ratio stays sqrt(2) at every permitted spacing.

Hermann Weyl's tile argument asks whether counting elementary steps recovers geometry at larger scales. Tobias Fritz (2013) makes a related difficulty precise for a class of periodic graphs (networks built from one repeating pattern): the attainable velocity set is a polytope with distinguished directions.

Causal set theory (spacetime as discrete events ordered by cause and effect), developed by Rafael Sorkin and others (Sorkin, 2003), pursues a different structure. Its continuum approximation combines causal order with a count related to volume. Poisson sprinkling, scattering points at random with a fixed average density, avoids the regular lattice's built-in preferred directions (Surya, 2019). Other discrete proposals make different commitments: Edward Fredkin (2003) advocates digital mechanics (the universe as a discrete computing process), Gerard 't Hooft (2016) investigates a cellular automaton (a grid of cells updated in steps by local rules) interpretation of quantum mechanics, Stephen Wolfram (2020) proposes rewriting rules on hypergraphs (networks whose links can join more than two points), and Carlo Rovelli and Lee Smolin (1995) derive discrete area and volume spectra, the possible measured values of area and volume. A discrete spectrum of geometric operators is not simply a lattice of physical locations, and Sabine Hossenfelder (2013) reviews the many forms a minimal length can take in quantum gravity, the sought theory uniting quantum physics and gravity. These programs can meet: a rewriting rule's causal graph, the record of which update depends on which, is itself a causal order, so one short rewriting rule whose causal graph matches a random causal set at large scales would carry both the computing proposals and causal-set theory. The search can also start

from the known laws, written as a program of discrete steps and shortened while its outputs still match.

A discreteness claim must state its level. A spatial lattice can carry real-valued fields and evolve in continuous time. A finite-dimensional quantum system can have a discrete measurement spectrum while retaining continuously many pure states, states as fully specified as quantum theory allows. Written out for a qubit, a two-level quantum system, $|\psi\rangle=\cos(\theta/2)|0\rangle+\exp(i\varphi)\sin(\theta/2)|1\rangle$, with continuously variable θ and φ, although measurement has only two outcomes. Two possible outcomes do not mean two possible states. The continuum of θ and φ is analysis, the description of every state a rule could reach. A history reaches its states one step at a time, so the values of θ and φ that occur form a countable set, and a finite set of quantum gates (basic operations of a quantum computer) comes arbitrarily close to every state of the continuum (Dawson and Nielsen, 2006).

4.2 What this paper contributes

This paper compares the two accounts under three obligations. First, distinguish the measure of a set from sums of singleton measures. Second, state which observables an effective representation reproduces, exactly or within a bound. Third, test the geometry and dynamics claimed for a physical realization, including failures that persist as a mesh is refined.

5. What would make discreteness a physical explanation?

A finite grid can approximate continuum measurements without proving that space is a grid. A physical alternative must specify its events and relations, show how measured distance and direction arise from them, and recover the predictions for which continuous physics already works. It must also reproduce

quantum interference: for two contributions of equal magnitude but opposite phase, their amplitudes cancel before probability is calculated.

A recovery example: waves on a spatial lattice

A lattice of sites reproduces continuum wave behavior at long wavelengths, within a stated error. Richard Feynman's lectures reach the continuum wave equation for an electron the same way, from amplitudes on a line of atoms (Feynman, Leighton and Sands, 1965, Ch. 16). Stated precisely, let u_j(t) be a displacement at site j of a one-dimensional lattice with spacing h, and take the nearest-neighbor dynamics (each site interacting only with the sites beside it)

u_j''=(c^2/h^2)(u_(j+1)-2u_j+u_(j-1)).

Substituting u_j(t)=exp[i(kjh-ωt)] gives

ω^2=(4c^2/h^2)sin^2(kh/2).

At long wavelengths the lattice relation reduces to the continuum one, with a correction set by the spacing. Written out, for |kh|<<1 it becomes ω^2=c^2k^2-c^2h^2k^4/12+O(h^4k^6). The leading term is the dispersion relation of the continuum wave equation, the link between a wave's frequency and its wavelength. A wave on the lattice therefore travels at slightly the wrong speed, and the error has a stated bound. In symbols, for 0<|kh|<=1 the relative phase-speed error obeys

0<=1-ω/(c|k|)<=(kh)^2/24.

Shrink h at fixed physical wavelength and this observable improves. Over a time T, the phase discrepancy is at most c|k|T(kh)^2/24 in this range; the agreement is not uniform over arbitrarily long times. Ben Schweizer and Florian Theil (2018) develop lattice-wave limits and long-time corrections.

In two dimensions the wave observable loses its direction dependence where the path metric keeps it. For the square-lattice analogue, $\omega^2=(4c^2/h^2)[\sin^2(k_xh/2)+\sin^2(k_yh/2)]$. Direction dependence remains at finite spacing but vanishes in the long-wavelength limit. This differs from the persistent sqrt(2) error of the shortest-path metric in Section 4.1 because the observable and dynamics are different.

This is a control calculation, not a new theory. The sites are discrete; the amplitudes and time parameter remain continuous here as analysis, describing many steps at once. It earns its place by being concrete: a declared dynamics yields a stated continuum observable with an error estimate.

6. Implications for mathematical monism

For mathematical monism, the view that reality is mathematical structure alone, the example sharpens what "structure" must include. A list of points together with their individual widths omits distances, measure, dynamics and observation relations. Complete structure cannot mean a pile of isolated constituents. Both discrete and continuous structures are mathematical; causality selects the discrete one for physical events.

Observer recovery

A complete candidate must grow observers like us. The universe starts simple, as observation shows, and on a measure where each bit of description halves the weight the probable worlds start simple too; minds are complex, so a history containing them must grow their organization from a simpler state. A short rule can generate an elaborate observer without becoming a long rule; output size and runtime are distinct from description length. Growth is a sequence of causal steps, one more reason the steps must have successors.

Conclusion

Everything that happens is discrete, matter, gravity, spacetime and space itself included, because causality takes steps and a continuum has none: between any two points lies a third, so no next state waits to be produced, and refining a grid supplies none. Change without a cause would be outside logic and reason, and an account that allowed it could no longer say why anything happens. The mathematics is not where the trouble lies. Zero point-lengths leave standard continuum measure theory intact, because a union of uncountably many points is not a countable sum of their lengths: the points do not cause the length, and continuum mathematics, amplitudes included, stays the analysis that describes many steps at once. Discreteness gives nothing a size: an element with a size would have a surface, and a surface is a continuum, so in every structure sizes come from relations among zero-width points. Of Zeno's two kinds of paradox, measure belongs to mathematics, which answers it, and motion belongs to physics, which needs steps.

The examples show why matching one average or shrinking a grid is not enough to recover all geometry. A discrete account succeeds by deriving the distances and dynamics we measure from steps that have successors.

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